INCH-POUND

MS75008F
W/AMENDMENT 1
1 August 2023
SUPERSEDING
MS75008F
31 August 2007

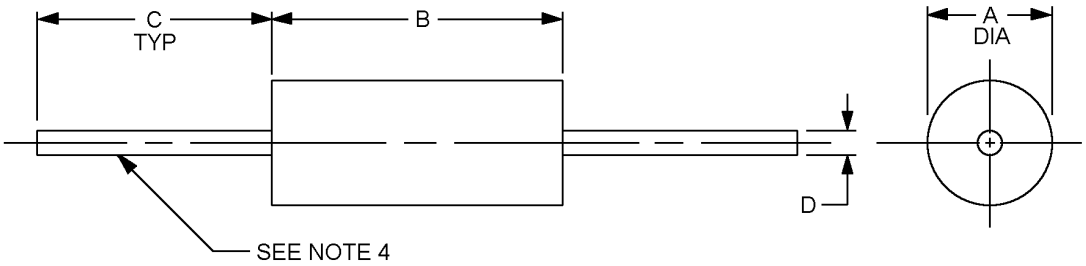
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, FIXED, SUBMINIATURE (PHENOLIC CORE), LT4K027 TO LT4K039, INCL

Inactive for new design,
Use MIL-PRF-39010/4

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



LTR	Minimum	Maximum
A	.156 (3.96)	.219 (5.56)
B	.406 (10.31)	.469 (11.91)
C	1.250 (31.75)	1.625 (41.28)
D	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These coils are intended to be mounted by their leads.
4. Solderable/weldable lead wire, number 22 AWG.

FIGURE 1. Dimensions and configuration.

AMSC N/A
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FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Weight: .95 gram, maximum.

Operating temperature range: -55° to +125°C.

Ambient temperature: + 90°C maximum.

Temperature rise: 35°C.

Altitude: 70,000 feet.

Shock, specified pulse: MIL-STD-202-213, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 1,000 V rms for a minimum of 60 seconds.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage 200 V rms for a minimum of 60 seconds.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Marking: Marking shall be as specified in MIL-PRF-15305 except that the marking shall be on the unit package or container.

Part or Identifying Number (PIN): MS75008 - (dash number from table I).

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TABLE 1. Electrical characteristics (initial).

Dash Number MS75008	Type designation	Inductance (μH)	Q Min.	Test Frequency (MHz)	SRF Minimum (MHz)	DC resistance max (Ohms)	Rated DC current, maximum (mA)
-21	LT4K027	0.15 ±20%	55	25	510	0.030	3,000
-22	LT4K028	0.22 ±20%	50	25	415	0.035	2,800
-23	LT4K029	0.33 ±20%	50	25	350	0.065	2,000
-24	LT4K030	0.47 ±20%	50	25	300	0.085	1,700
-25	LT4K031	0.56 ±10%	50	25	270	0.125	1,450
-26	LT4K032	0.68 ±10%	45	25	250	0.150	1,300
-27	LT4K033	0.82 ±10%	40	25	210	0.205	1,100
-28	LT4K034	1.00 ±10%	40	25	200	0.290	930
-29	LT4K035	1.20 ±10%	30	7.9	180	0.400	785
-30	LT4K036	1.50 ±10%	30	7.9	170	0.485	700
-31	LT4K037	1.80 ±10%	30	7.9	150	0.740	580
-32	LT4K038	2.20 ±10%	30	7.9	140	0.970	505
-33	LT4K039	2.70 ±10%	30	7.9	120	1.20	460

TABLE II. Electrical characteristics (final).

Inspection group	Allowable variation from Initial measurement	Allowable variation from Initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2	---	---	-10
Group III	±5	±(3% +.001 ohm)	-8	-10
Group IV	±5	±(2% +.001 ohm)	-10	-15
Conformance inspection group C				
Subgroup I	±2	---	---	-10
Subgroup II	±5	±(2% +.001 ohm)	-10	-15
Subgroup III	±5	±(3% +.001 ohm)	-8	-10

Application notes:

1. The polarization voltage during moisture the moisture resistance tests is applied with the positive lead Connected to the coil terminals tied together, and the negative lead connected to the metal strap.
2. Former MS part numbers MS75008-34 through MS75008-45 have been superseded by MS75101-1 through MS75101-12, respectively.

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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-15305, this document references the following:

MIL-STD-202-105 MIL-STD-202-213 MIL-STD-202-301 MS75101

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2023-022

Review activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

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